

Impact of climate change interventions on pear production in

Belgium

NL_Draft

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R Packages

```
library(ggplot2)
library(MASS)
library(ggplot2)
library(tidyverse)
library(scales)
library(factoextra)
```

Data Import

```
pear_data <- read.csv("Data/G12.size.data.csv")
outcome_data <- read.csv("Data/G12.outcome.data.csv")
data <- read.csv("Data/G12.soil.data.csv")
```

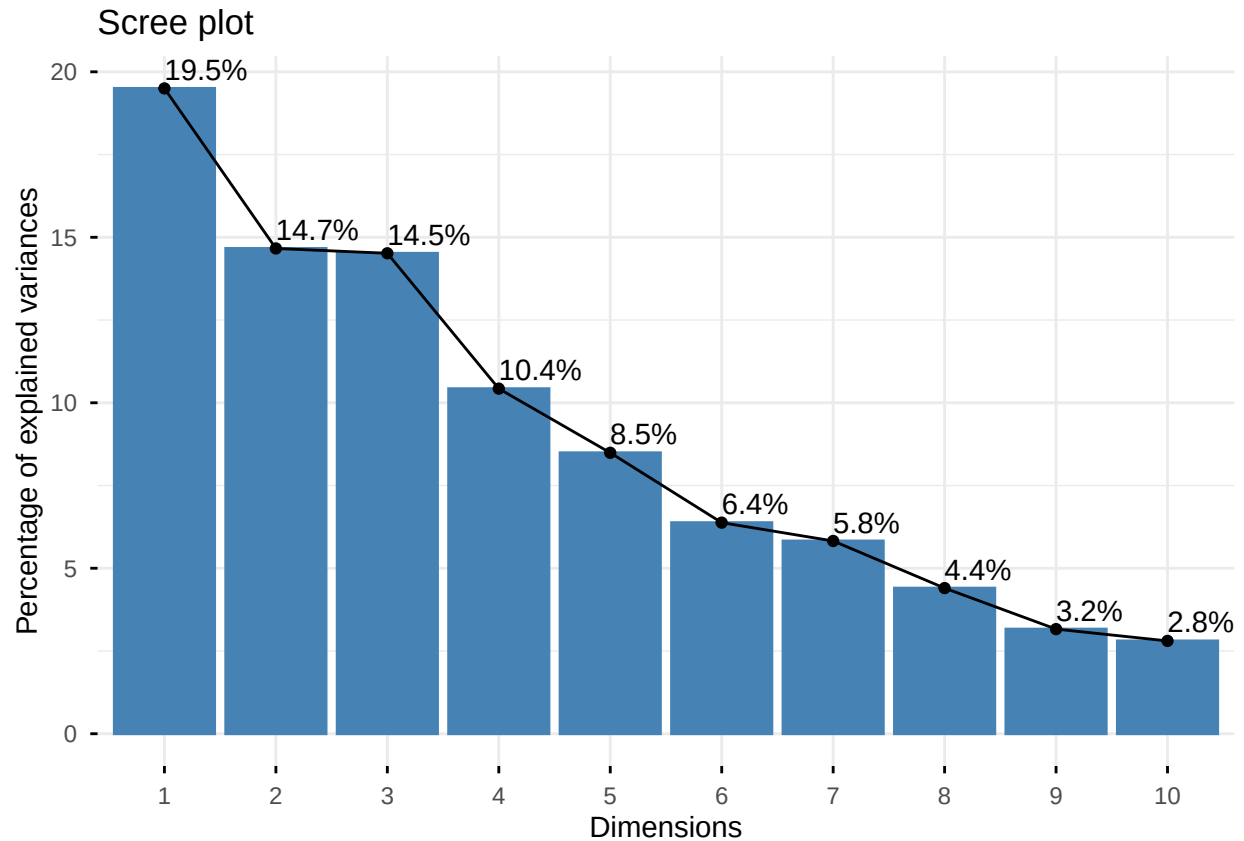
Exploration of soil condition features using multivariate methods

To explore patterns in soil condition features and gain insights into the effect of climate on pear production, both Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) were applied.

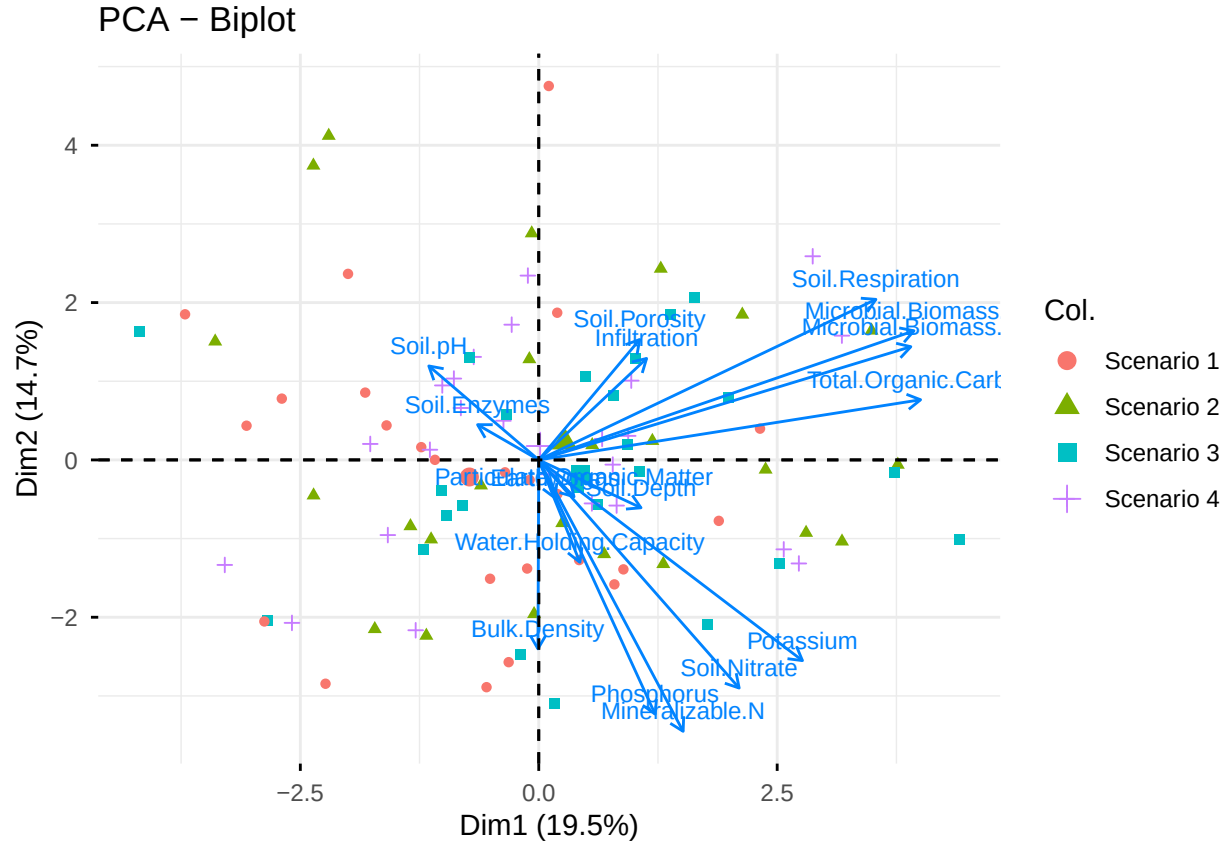
Principal Component Analysis (PCA)

PCA was used as an unsupervised method to investigate the underlying structure of the soil variables without considering the climate grouping. Before performing PCA, all soil variables were standardized to ensure comparability, as they were measured on different scales. The PCA reduces the dimensionality of the data by transforming the original variables into a set of uncorrelated principal components that capture the maximum variance. Although the common rule is the chosen principle components should be able to explain at least 80% of the total variance, due to visualization constraints, only the first two components were shown in a two-dimensional plot in the report.

```
cols_to_remove <- c("tree_ID", "location", "ecotr_id", "tree_id", "species", "climate")
soil_data <- select(data, -all_of(cols_to_remove))
pca_data <- prcomp(soil_data, center = TRUE, scale = TRUE)
fviz_eig(pca_data, addlabels = TRUE)
```



```
fviz_pca_biplot(pca_data, repel = FALSE,
  geom.ind = "point",
  col.var = "#0083FF",
  col.ind = data$climate,
  labelsize = 3
)
```



The scree plot shows that the first principal component (PC1) explains 19.5% of the total variance, while the second principal component (PC2) explains 14.7%, resulting in a combined explained variance of approximately 34.2%. This indicates that the variability in soil features is distributed across multiple components, with no single dominant dimension.

The PCA biplot was used to explore whether soil conditions differ across climate scenarios. However, no clear clustering or separation of the four climate scenarios is observed on the PC1–PC2 plane. The observations from different scenarios are widely overlapping and appear to be randomly dispersed.

Moreover, no clear directional separation can be identified from the loading vectors. The projections of observations onto individual variable vectors do not show consistent or distinguishable patterns across climate scenarios. This suggests that none of the soil features, as represented in the first two principal components, provides a strong indication of differences between the groups.

Overall, the PCA results indicate that, based on the variation captured by PC1 and PC2, there is no strong evidence that soil conditions differ substantially across climate scenarios. This implies that if differences exist, they are likely subtle or distributed across higher-dimensional structures not captured by the first two principal components.

Linear Discriminant Analysis (LDA)

To further assess whether soil conditions (together with pear production) can discriminate between climate scenarios, LDA was performed. In this analysis, the climate scenario was used as the grouping variable, while soil features and the number of pears were included as predictors. LDA aims to find linear combinations of variables that maximize the separation between predefined groups, thereby helping assess whether different climate scenarios are associated with distinct profiles of soil conditions and pear production.

```

pear_count <- summarise(
  group_by(outcome_data, location, ecotr_id, tree_id),
  pears = n()
)
soil_with_pears <- left_join(
  data,
  pear_count,
  by = c("location", "ecotr_id", "tree_id")
)
predictors <- select(
  soil_with_pears,
  -tree_ID,
  -location,
  -ecotr_id,
  -tree_id,
  -species,
  -climate
)
group <- as.factor(data$climate)
predictors_scaled <- scale(predictors)
lda_model <- lda(group ~., data = as.data.frame(predictors_scaled))
lda_result <- predict(lda_model)
lda_scores <- as.data.frame(lda_result$x)
lda_scores$climate <- data$climate

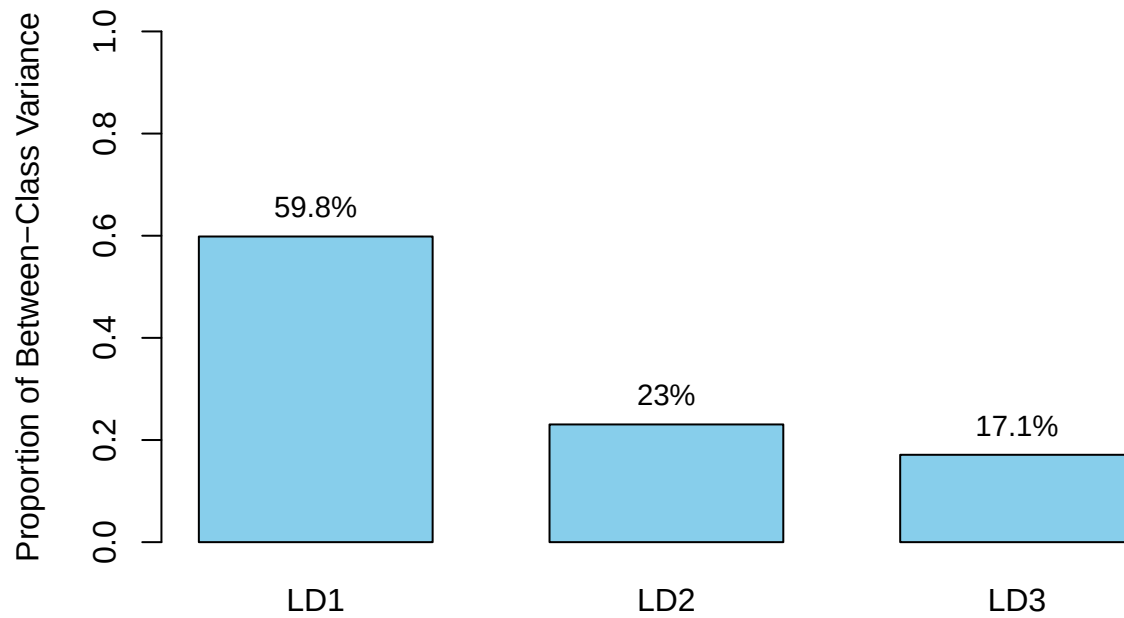
# Extract proportion of between-class variance
lda_var <- lda_model$svd^2
lda_var_ratio <- lda_var / sum(lda_var)

# Barplot of eigenvalues (proportion of variance)
midpoints <- barplot(
  lda_var_ratio,
  names.arg = paste0("LD", 1:length(lda_var_ratio)),
  col = "skyblue",
  ylab = "Proportion of Between-Class Variance",
  main = "LDA: Eigenvalues",
  ylim = c(0, 1),
  width = 0.3,
  space = 0.5
)

text(
  x = midpoints,
  y = lda_var_ratio,
  labels = paste0(round(lda_var_ratio * 100, 1), "%"),
  pos = 3,
  cex = 0.9
)

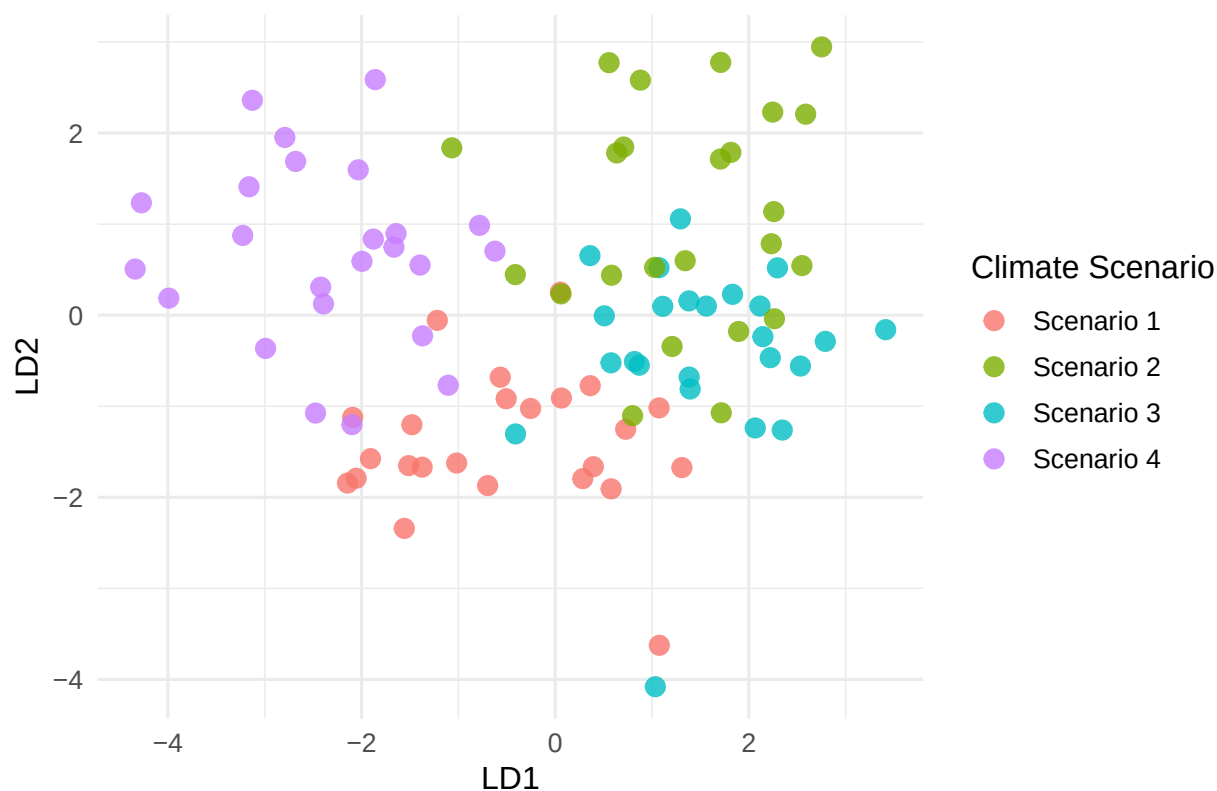
```

LDA: Eigenvalues



```
# LDA scatter plot
ggplot(lda_scores, aes(x = LD1, y = LD2, color = as.factor(climate))) +
  geom_point(size = 3, alpha = 0.8) +
  labs(
    title = "LDA Scatter Plot (LD1 vs LD2)",
    x = "LD1",
    y = "LD2",
    color = "Climate Scenario"
  ) +
  theme_minimal(base_size = 12)
```

LDA Scatter Plot (LD1 vs LD2)



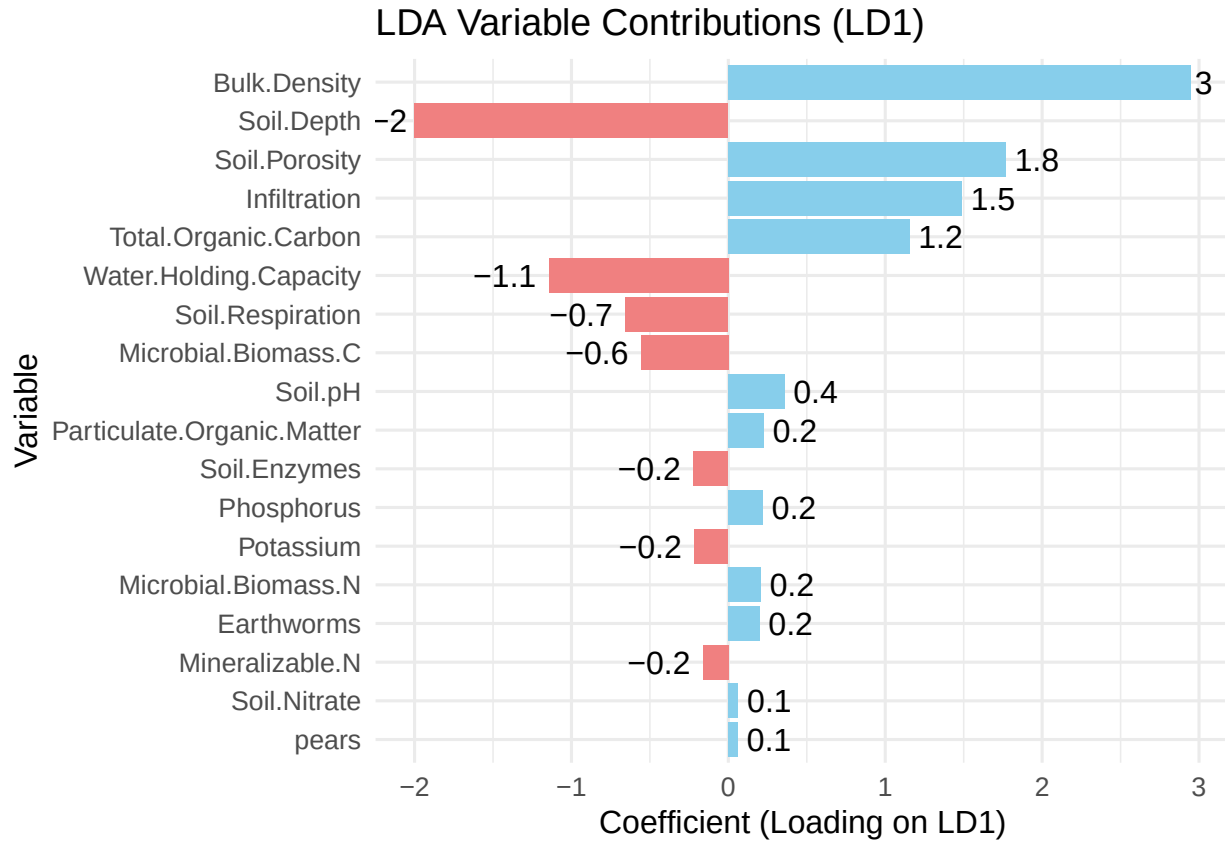
```
# LD1 contributions
ld1_loadings <- lda_model$scaling[, 1]

loading_df <- data.frame(
  Variable = names(ld1_loadings),
  LD1 = ld1_loadings
)

loading_df$Color <- ifelse(loading_df$LD1 < 0, "lightcoral", "skyblue")

loading_df <- mutate(loading_df, Variable = fct_reorder(Variable, abs(LD1)))

ggplot(loading_df, aes(x = Variable, y = LD1, fill = Color)) +
  geom_col() +
  geom_text(
    aes(label = round(LD1, 1)),
    hjust = ifelse(loading_df$LD1 > 0, -0.2, 1.2)
  ) +
  scale_fill_identity() +
  coord_flip() +
  labs(
    title = "LDA Variable Contributions (LD1)",
    x = "Variable",
    y = "Coefficient (Loading on LD1)"
  ) +
  theme_minimal(base_size = 12)
```



The eigenvalue barplot shows that the first linear discriminant (LD1) explains approximately 59.8% of the between-class variance, followed by LD2 (23%) and LD3 (17.1%). This indicates that the majority of the group separation can be captured by LD1, with additional contributions from higher dimensions.

The LDA scatter plot (LD1 vs LD2) shows a clearer separation between climate scenarios compared to PCA, particularly along LD1. Scenario 4 is mainly located on the negative side of LD1, while Scenario 2 and Scenario 3 tend to appear on the positive side, with Scenario 1 positioned in between. This suggests that soil features provide meaningful discrimination between climate conditions when group information is incorporated.

The coefficients of LD1 indicate which variables contribute most to the separation. Variables such as Bulk Density, Soil Porosity, Infiltration, and Total Organic Carbon have relatively strong positive contributions, while Soil Depth and Water Holding Capacity contribute negatively. This implies that climate scenarios on the positive side of LD1 are associated with soils that are more porous and biologically active, whereas those on the negative side are characterized by deeper soils with higher water retention capacity.

Importantly, the variable pear production (number of pears) has a very small coefficient, indicating that it contributes minimally to the discrimination between climate scenarios. This suggests that the observed separation is primarily driven by differences in soil conditions rather than direct differences in pear yield.

Conclusion

Overall, the results suggest that soil conditions may vary slightly across climate scenarios, as indicated by the separation observed in LDA. However, these differences are not dominant in the overall variability of the data, as PCA does not show clear separation. In contrast, pear production shows negligible contribution to the discrimination, indicating that it does not differ substantially across climate scenarios. Hence, there is

not enough evidence from this multivariate analysis to conclude that climate scenarios have a meaningful impact on pear production.